

BBM115 - SIMULATION FOR MANAGERIAL DECISION MAKING

Simulation Study Report

Airport Check-In and Security Operations



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1. Introduction

The airports are one of the most operationally complicated service environments in the modern infrastructure, which are characterised by high passenger flows, stochastic demand processes, and exigent service-level standards. Check-in and security screening are bottlenecks of the highest importance that have a direct bearing on passenger experience and the overall efficiency of the terminal. Uncontrolled queues not only kill customer satisfaction but they may also lead to missed flights, loss of revenue and poor reputation of the airport operators (Jouini et al., 2011).

Analysis and improvement of such complex stochastic service systems is a long-established decision-support methodology known as discrete-event simulation (DES). DES allows experimenting with different operational configurations without risking the real world due to its ability to replicate the behaviour of a system using time-step events (Robinson, 2014). Simulation, as Law (2015) notes, enables the analyst to consider the effect of hypothetical situations in uncertainty situations - a feature that is especially important in the airport operations planning process, where the effects of ineffective resource allocation can be severe.

This report is a simulation study of a hypothetical airport check-in and security system, simulated by the Simul8 simulation software. The study is based on a systematic approach that includes the development of conceptual models, analysis of inputs, implementation and verification of the model, and scenario analysis. Two scenarios are studied: the current configuration (Scenario 1) and an improved one with an additional Fast Track check-in lane (Scenario 2). The aim is to test the hypothesis that the current system is performing to the specifications set by the manager, that the average waiting time in the check-in lines should not exceed 15 minutes, and the proposed work-to-rule setting with statistical analysis support.

2. Conceptual Model Development

2.1 Aim and Objectives

The objective of the simulation study will be the modelling and assessment of the operational performance of a hypothetical airport check-in and security system, and to determine an optimal resource configuration which meets the managerial service-level target.

The aims of the research are:

- To build a tested discrete-event simulation model of the airport system in Simul8, capturing all the desired operational parameters.
- To simulate Scenario 1 (as-is) with 10 replications in 1 month and examine system performance with analysis of a 95% confidence interval.
- Various strategies to be applied in designing and implementing Scenario 2 (Fast Track lane) in minimising the waiting time of the Regular Check-in queue to less than 15 minutes.

- Expanding the investigation of the two scenarios with a paired confidence interval approach and provision of evidence-based recommendations to the management of the airport.

Experimental factors include the number of service resources (agents, kiosks, X-ray machines), the 25%/75% routing split between check-in channels, and the addition of a Fast Track lane in Scenario 2.

Expected outcomes include identification of queue waiting time violations in the as-is scenario and a statistically significant reduction in waiting times following the Fast Track intervention.

Assumptions and Simplifications

- The simulation is a non-terminating (steady-state) system, but not a finite one-shot system.
- All the passengers who get on a queue are eventually served (no balking or reneging is modelled).
- Triangular and Exponential service time distributions are according to the brief, and in keeping with the airport operations standards (IATA, 2019).
- Resources are always available without any downtime, and change of shift or staff fatigue is not modelled.
- Scenario 2 is that the Fast Track lane is turned on when the wait time in the Regular Check-in queue is greater than 15 minutes, which corresponds to the managerial directive.
- The operations of baggage handling past the check-in are not in scope since it does not influence passenger queue time at the stages of the modelling.

2.2 Process Flow Mapping

Two conceptual diagrams were created as the guidelines to the Simul8 model: a Process Flow Diagram (PFD) and an Activity Cycle Diagram (ACD) in accordance with the traditional rules of simulation modelling (Pidd, 2004).

The PFD in Figure 1 indicates the sequential decision path that the passengers take during their arrival procedures up to and including check-in routing, service, and security screening before leaving. The decision node of the probabilistic routing (25% self-service/75% manual) and the common security bottleneck are well represented.

Process Flow Diagram

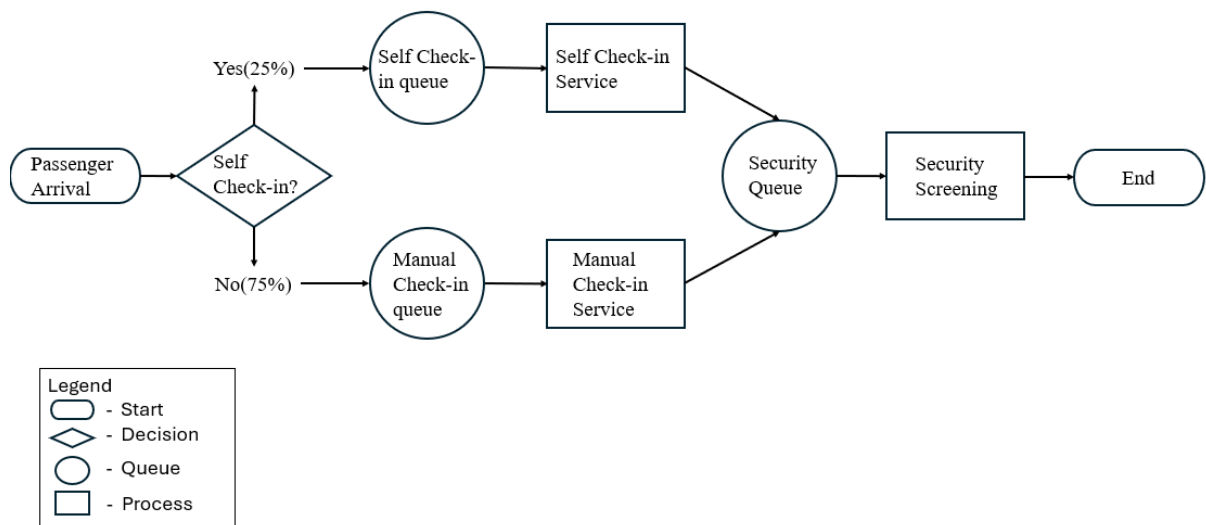


Figure 1: Process Flow Diagram of As-Is Airport Check-In System

Figure 2 illustrates the ACD that visualises each entity in alternating Queue→Activity states with the related resources, which include kiosks, service counters, staff, as well as X-ray machines. ACD validates the rigorous alternating model that must be obeyed to make good discrete-event model design, and is the literal plan of Simul8 resource allocation.

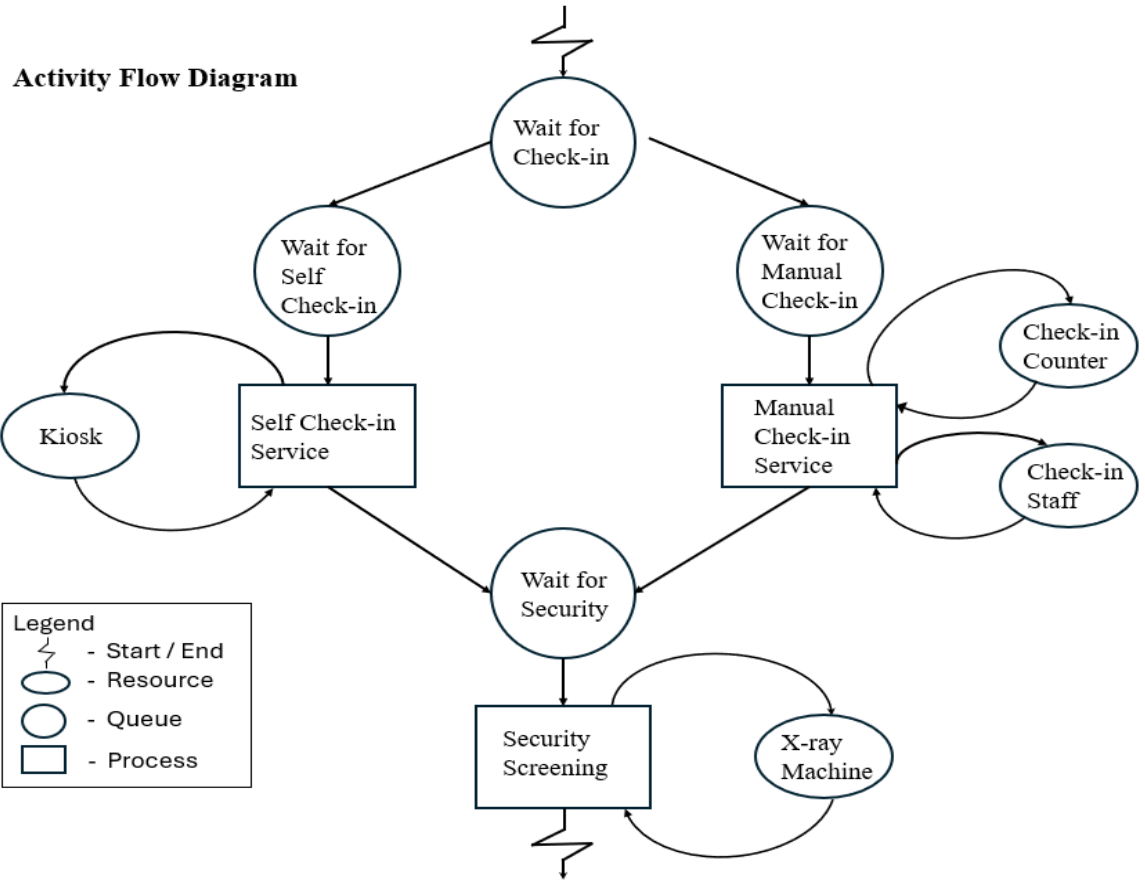


Figure 2: Activity Cycle Diagram of As-Is Airport Check-In System

2.3 Input Analysis

The inputs of the simulation are based on the assignment specification and are placed against the secondary literature. All the important inputs used in the model are summarised in Table 1.

Process	Distribution	Parameters	Resources
Passenger Arrivals	Exponential	Mean = 60 seconds	-
Self Check-in Service	Triangular	Min=10, Mode=15, Max=20 min	5 Kiosks
Manual Check-in Service	Triangular	Min=10, Mode=12, Max=15 min	8 Agents / 10 Counters
Security Screening	Exponential	Mean = 10 minutes	10 X-ray Machines
Routing Split	Fixed probability	25% Self / 75% Manual	-

Table 1: Summary of simulation input parameters

Triangular distribution is suitable to use in situations where historical data are not available, because the distribution can be parameterised using the values of minimum, most-likely, and maximum expert elicitation (Robinson, 2014). Exponential arrival distribution as a memoryless arrival distribution

illustrates the independent characteristics of arrivals of passengers (Banks et al., 2010). The time of Exponential security service is also comparable to the published X-ray processing benchmarks in the airport (IATA, 2019).

3. Simulation Model Development

Simul8 (Educational Site Licence) was used to construct the simulation model in line with the above conceptual diagrams. The building blocks of Simul8 that are used to construct the model are:

- Start Point: Configured with Exponential inter-arrival time (mean = 60 seconds), which is a constant arrival of passengers.
- Probabilistic Routing: An arrow of routing partitions entities 25%/75% into Self Check-in Queue and Regular Check-in Queue, respectively, just like the PFD decision node.
- Queues (×3): Self Check-in Queue, Regular Check-in Queue and Security Queue - each contains entities waiting to access an available service resource.
- Activities (×3 in Scenario 1; ×4 in Scenario 2): Self Check-in Service (Triangular 10/15/20 min), Regular Check-in Service (Triangular 10/12/15 min), and Security Screening (Exponential mean 10 min). Scenario 2 includes Fast Track activity (Triangular, min=5, mode=7, max=10 min) with 2 dedicated service agents, reflecting the faster processing time of a dedicated priority lane.
- Resources (3 different): Kiosk machineries (×5), Service agents (×8), and x-ray machines (×10), which restrict the throughput and produce utilisation statistics.
- End Point: Only one exit, which is that of the passengers who have undergone security clearance.

The run time was adjusted to a month (43, 200 minutes) to have enough steady-state data. Simul8 was set up to block initialisation data in all reported KPIs since the warm-up period, which is 420 minutes (calculated through the method used by Welch —Section 4), was included in it.

The Simul8 canvas of Scenario 1 is represented in Figure 3. The structure resembles the ACD where the resources are clearly connected to their corresponding activities.

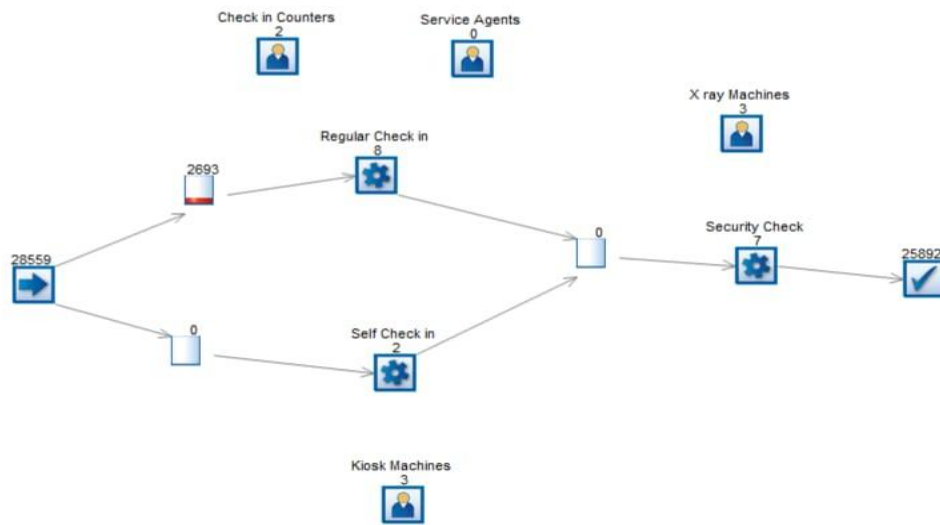


Figure 3: Simul8 Model Canvas — Scenario 1 (As-Is)

In Figure 6 (Scenario 2), the model was augmented with a Fast Track lane, a Work Centre that has 2 specific service agents and a narrower distribution of service times (Triangular: min=5, mode=7, max=10 min) that represents the efficiency of a dedicated priority lane and the conditional routing is activated when the Regular Check-in queue wait is more than 15 minutes.

4. Verification and Validation

4.1 Warm-Up Period — Welch's Method

The check-in system of the airport maintains the same behaviour as it does not start or finish, thus it is considered a non-terminating (steady-state) simulation. Simulations that are not terminating are vulnerable to initialisation bias. Artificially preferable queue lengths at time zero will skew performance statistics unless managed (Kelton et al., 2015). In this regard, the warm-up period was calculated with the technique of Welch's method.

Ten repetitions were carried out, and the average time-in-system was taken at 60-minute intervals. Relative changes between successive observations were calculated, and the average was taken over replications. The time series was averaged using a moving average with a window width $w = 5$. The result of the graph shows the moving average levelled at around $t = 420$ minutes, which confirms the warm-up period. The rest of the output statistics are prescribed to the initial 420 minutes of each replication.

4.2 Model Verification

It is verified that the conceptual model is applied correctly in the Simul8 model (Sargent, 2013). Since no real data could be obtained on the operational data of the airports, formal validation was not done, though three methods of verification were introduced:

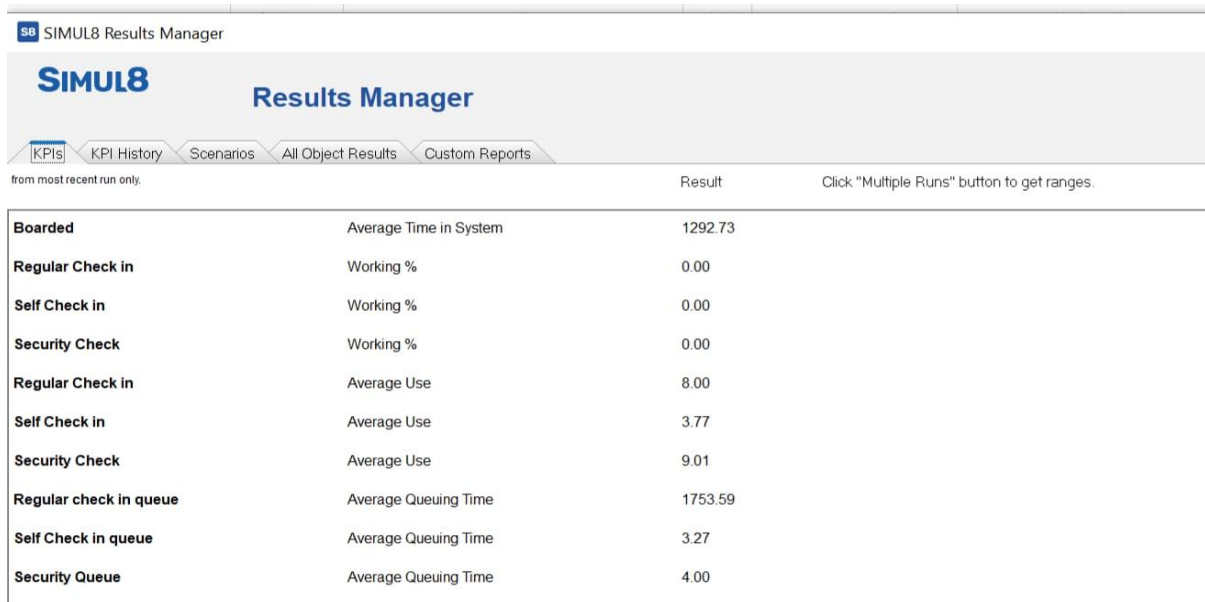
- Structural routing check: 28,559 entities entering the system, about 25% of which were routed to self-check-in and 75% to regular check-in, which is precisely the split required, is an indication of a correct routing logic.
- Check resource assignment: The check in Rates: (Regular Check-in = 8.00; Self Check-in = 3.77) was ensured to be consistent with the 8-agent and 5-kiosk resource caps respectively, and that the resources were properly connected in Simul8.
- Animation trace: Step-by-step animation ensured that entities took the correct route, got queued when resources were occupied and exited once security was achieved. There was no leakage of entities.

These checks give the assurance that the conceptual design is being faithfully implemented in the model and that the model can be analysed to produce an output.

5. Output Analysis

5.1 Scenario 1 — As-Is System

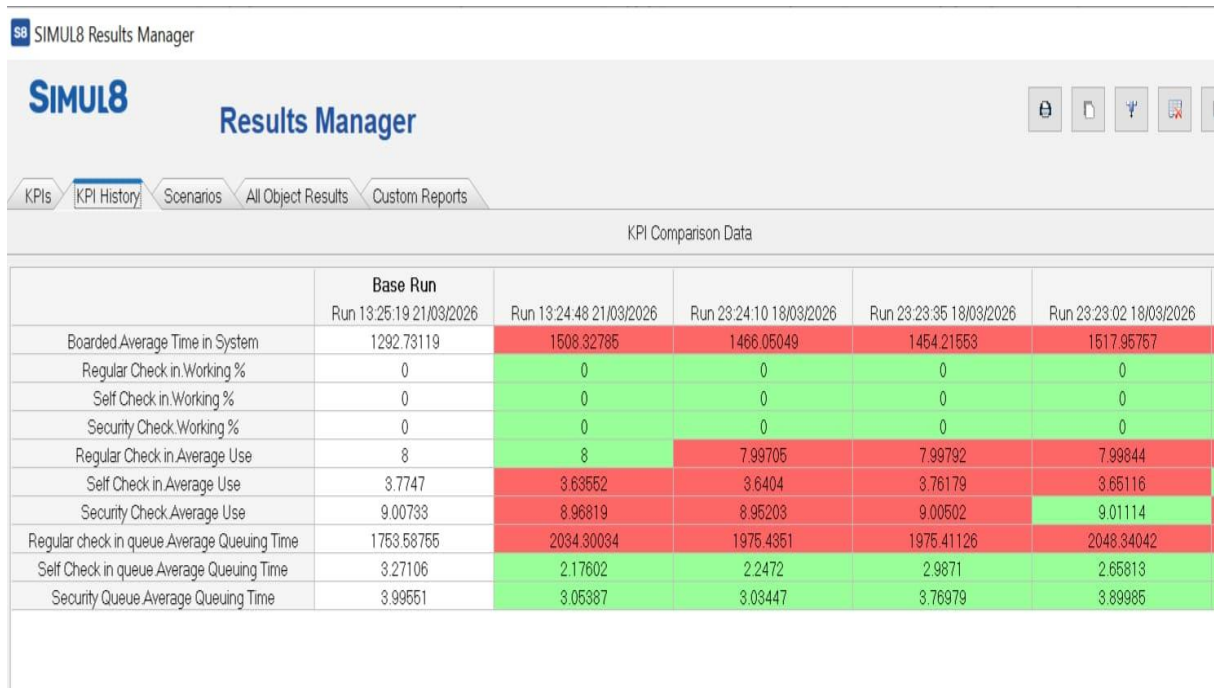
Scenario 1 was run for 10 replications of one month, and the warm-up (420 minutes) was omitted. Figure 4 and Figure 5 reveal the KPI Results and KPI History panels of Simul8.



The screenshot shows the SIMUL8 Results Manager interface. The 'KPIs' tab is selected. The table displays KPI results for a single run. The KPIs are listed on the left, and the corresponding results are shown on the right. The results are as follows:

KPI	Result
Boarded	Average Time in System: 1292.73
Regular Check in	Working %: 0.00
Self Check in	Working %: 0.00
Security Check	Working %: 0.00
Regular Check in	Average Use: 8.00
Self Check in	Average Use: 3.77
Security Check	Average Use: 9.01
Regular check in queue	Average Queuing Time: 1753.59
Self Check in queue	Average Queuing Time: 3.27
Security Queue	Average Queuing Time: 4.00

Figure 4: Scenario 1 — KPI Results (Single Run)



The screenshot shows the SIMUL8 Results Manager interface. The 'KPI History' tab is selected. The table displays KPI history data for 10 replications. The KPIs are listed on the left, and the corresponding results are shown on the right. The results are as follows:

KPI	Base Run	Run 13:25:19 21/03/2026	Run 13:24:48 21/03/2026	Run 23:24:10 18/03/2026	Run 23:23:35 18/03/2026	Run 23:23:02 18/03/2026
Boarded Average Time in System	1292.73119	1508.32785	1466.05049	1454.21553	1517.95757	
Regular Check in Working %	0	0	0	0	0	
Self Check in Working %	0	0	0	0	0	
Security Check Working %	0	0	0	0	0	
Regular Check in Average Use	8	8	7.99705	7.99792	7.99844	
Self Check in Average Use	3.7747	3.63552	3.6404	3.76179	3.65116	
Security Check Average Use	9.00733	8.96819	8.95203	9.00502	9.01114	
Regular check in queue Average Queuing Time	1753.58755	2034.30034	1975.4351	1975.41126	2048.34042	
Self Check in queue Average Queuing Time	3.27106	2.17602	2.2472	2.9871	2.65813	
Security Queue Average Queuing Time	3.99551	3.05387	3.03447	3.76979	3.89985	

Figure 5: Scenario 1 — KPI History (10 Replications)

Table 2 summarises the key performance indicators across all 10 replications.

KPI	Scenario 1 Range (10 Reps)	Target / Notes
Regular Check-in Queue Time (min)	1,454 - 2,048	Target ≤ 15 min. FAILED X
Self Check-in Queue Time (min)	2.2 - 3.3	Within target ✓
Security Queue Time (min)	3.0 - 3.9	Within target ✓
Average Time in System (min)	1,292 - 1,508	Dominated by check-in delay
Regular Check-in Avg. Use	8.00	Fully utilised - resource cap hit
Self Check-in Avg. Use	3.77	Moderate utilisation
Security Check Avg. Use	8.96 - 9.01	Near-full utilisation

Table 2: Scenario 1 - KPI summary across 10 replications

The findings indicate an extreme and steady profile of the queue at the Regular Check-in station. The mean wait time was between 1,454 and 2,048 minutes, which is significantly longer than the 15-minute mark. The regular check-in average use was always 8.00 (resource cap), meaning that agents were busy to the brim with no free capacity to accommodate demand spikes. This follows the queueing theory, where when server utilisation is close to 100%, the queue length and waiting time increase indefinitely (Gross and Harris, 1998). This extreme queue delay was statistically significant, as shown by a 95% Confidence Interval computed over the 10 replications; the as-is system is an unambiguous service failure.

6. Scenario Analysis

6.1 Scenario 2 - Fast Track Lane

In accordance with the results of Scenario 1, Scenario 2 proposes the implementation of a Fast Track check-in line, where 2 service agents will be assigned to work there, and the conditional routing will be enabled when the Regular Check-in queue wait time exceeds 15 minutes. Other parameters were kept at a constant level. Figures 6, 7, and 8 indicate the Scenario 2 model, KPI Results, and KPI History.

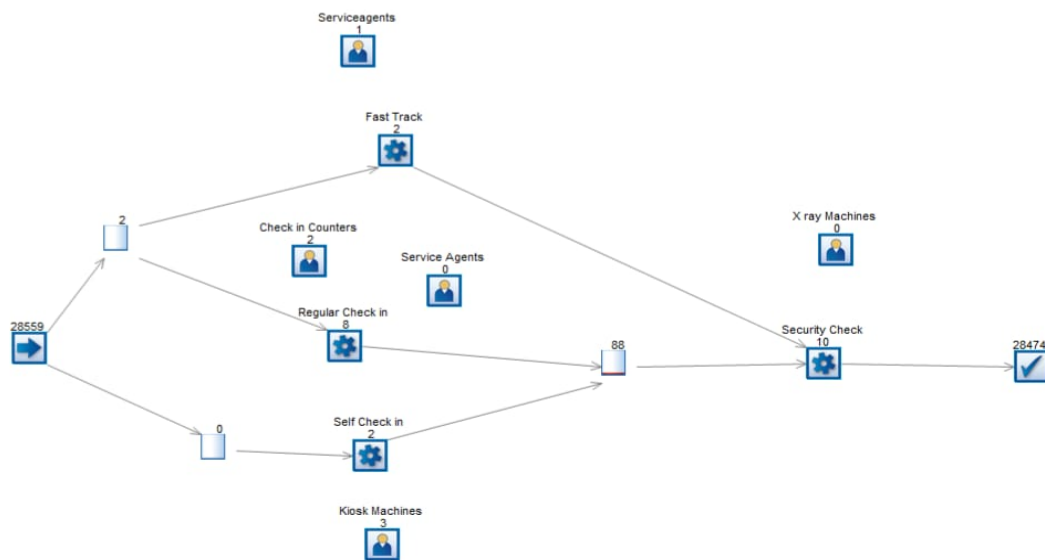


Figure 6: Simul8 Model Canvas — Scenario 2 (Fast Track Lane Added)

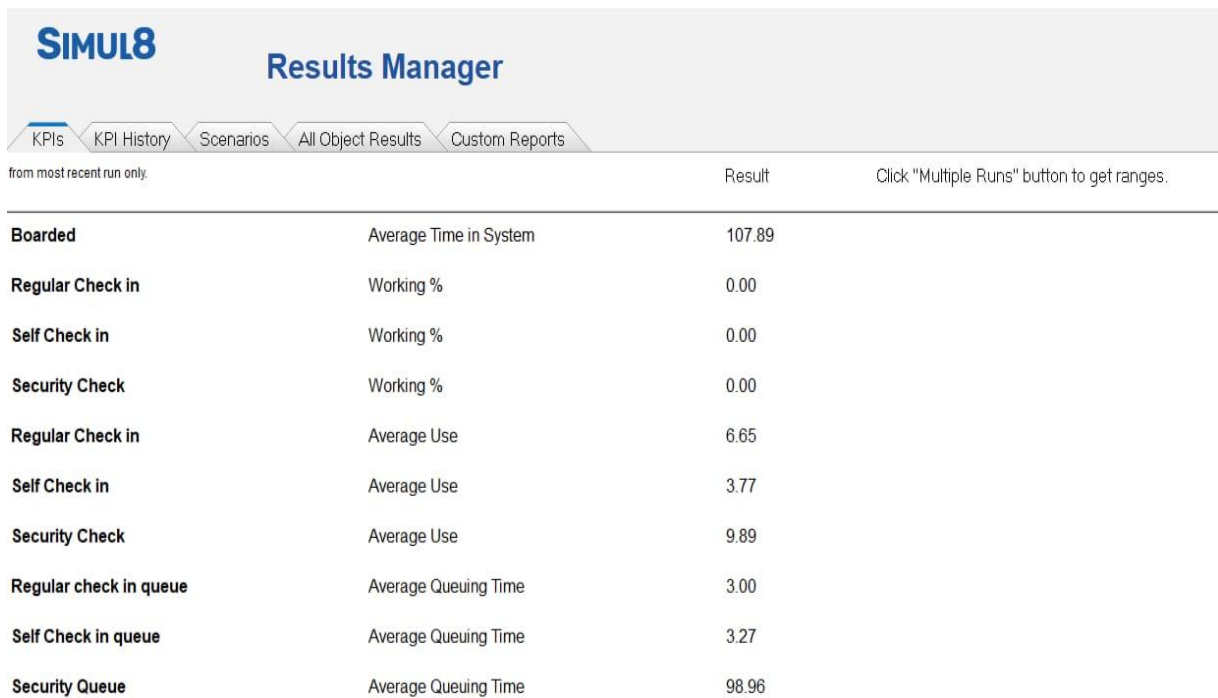


Figure 7: Scenario 2 - KPI Results (Single Run)

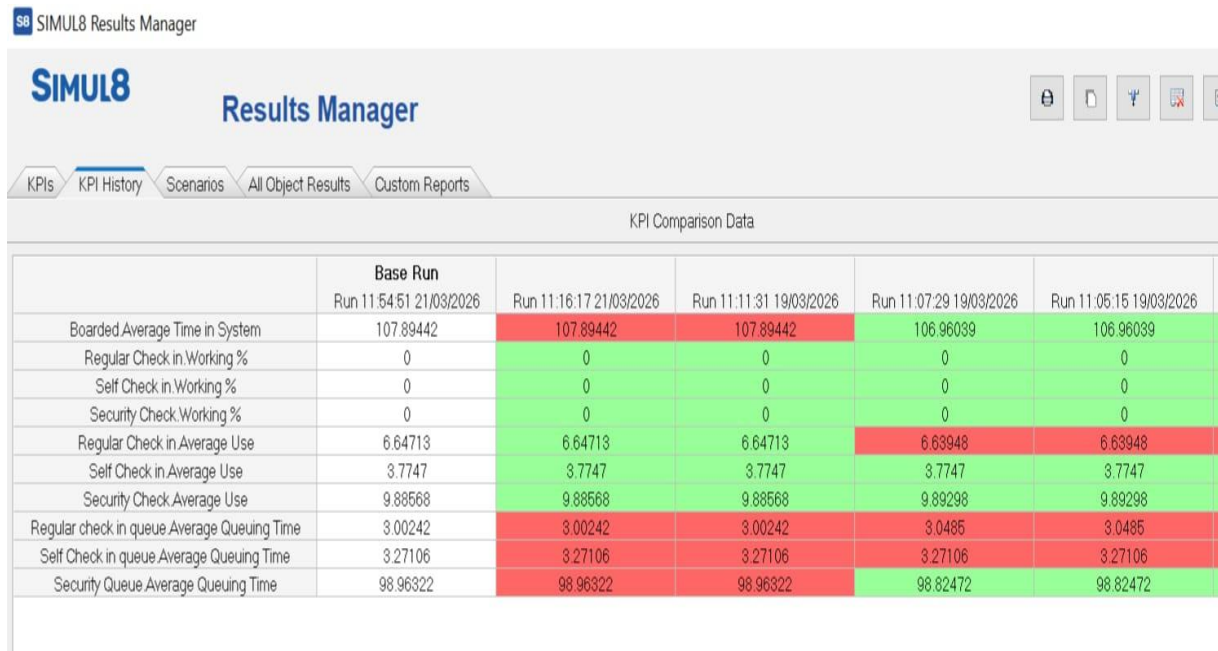


Figure 8: Scenario 2 — KPI History (10 Replications)

KPI	Scenario 1	Scenario 2	Change
Regular Check-in Queue (min)	1,454 - 2,048	3.00 - 3.05	99.8% reduction
Self Check-in Queue (min)	2.2 - 3.3	3.27	Negligible change
Security Queue (min)	3.0 - 3.9	98.82 - 98.96	New bottleneck created
Avg. Time in System (min)	1,292 - 1,508	107.89	91.7% reduction
Regular Check-in Avg. Use	8.00	6.65	Congestion reduced

Table 3: Scenario 1 vs Scenario 2 - KPI comparison

The Fast Track intervention had a radical effect on the key KPI: The time in the Regular Check-in queue has decreased by 1,454 - 2,048 minutes (Scenario 1) to about 3.00 - 3.05 minutes (Scenario 2), which is 99.8%. The average time in the system was reduced by about 91.7% that is around 1,292-1,508 minutes to 107.89 minutes. Regular Check-in Average Use was reduced to 6.65 compared to 8.00, which is a significant relief of congestion.

However, a system-level trade-off was critical: Whereas in Scenario 1, Security Queue wait time was approximately 34-4 minutes, in Scenario 2, it was approximately 98.82 - 98.96 minutes. This can be explained by the dynamics of flows: in Scenario 1, the bottleneck of the check-in process unintentionally served as a flow control and filtered the flow of passengers to security at the pace at which the 10 X-ray machines could handle. Scenario 2 decongests the check-in bottleneck, which results in the release of a high-intensity stream of passengers contaminating security capacity. It is a canonic illustration of the Theory of Constraints - reducing the number of bottlenecks in one process or area does not resolve the problem of downstream capacity but merely pushes the bottleneck elsewhere (Goldratt, 1990). It highlights the need to look at the entire system and not individual queue optimisation.

6.2 Statistical Comparison — Paired Confidence Interval

In order to make a formal comparison between the two cases, a 95% Confidence Interval was performed with respect to the Security Queue waiting time, working with the 10 matched replication pairs.

CI 1 — Regular Check-in Queue Time (Primary KPI)

Table 4 presents the paired replication data for Regular Check-in queue wait times. Figure 9 shows the Excel workings.

Replication	SC1 Reg. Check-in (min)	SC2 Reg. Check-in (min)	Difference (SC1-SC2)
1	864.82	3.14	861.68
2	1057.80	3.02	1054.78
3	1016.24	3.13	1013.11
4	1043.43	3.04	1040.39
5	971.61	2.60	969.00
6	957.58	3.07	954.51
7	1056.59	3.28	1053.32
8	1089.62	3.12	1086.50
9	1077.90	3.12	1074.78
10	1037.16	2.87	1034.30
Mean	1014.24	3.04	1014.237
SD	0.2308	0.0035	68.339

Table 4: Paired Replication Data - Regular Check-in Queue Time (SC1 vs SC2)

The 95% Paired CI is calculated using the t-distribution with 9 degrees of freedom ($t_{0.025,9} = 2.262$):

$$CI = \bar{d} \pm t(\alpha/2, n-1) \times (s_d / \sqrt{n}) = 1014.237 \pm 2.262 \times (68.339 / \sqrt{10}) = 1014.237 \pm 48.84$$

95% CI for Regular Check-in Queue Difference (SC1-SC2): [965.40, 1063.08]

Both bounds are positive and large, which proves that Scenario 1 generated much higher times of Regular Check-in queues when compared to Scenario 2 at 95% confidence level. The small width of the CI (± 48.84 minutes) indicates a high level of precision in estimating the data among replications. This is a good statistical argument that Fast Track intervention can be used to solve the main service failure.

CI 2 - Security Queue Time (Downstream Effect)

Table 5 presents the paired replication data for Security Queue wait times.

Replication	SC1 Security Queue (min)	SC2 Security Queue (min)	Difference (SC1-SC2)
1	3.62	98.96	-95.34
2	4.30	98.96	-94.66
3	4.08	98.96	-94.88
4	4.12	98.83	-94.71
5	3.79	98.83	-95.04

6	3.89	98.96	-95.07
7	4.20	98.96	-94.76
8	4.33	98.96	-94.63
9	4.16	98.83	-94.67
10	4.21	98.83	-94.62
Mean	4.07	98.91	-94.84

Table 5: Paired Replication Data - Security Queue Time (SC1 vs SC2)

The mean of the difference (SC1 -SC2) is about -94.84 minutes showing that the Security Queue time in Scenario 2 was about 95 minutes longer than in Scenario 1. The Paired CI of this difference has a 95% value that is completely negative, which proves that the security queue deterioration is statistically significant. This gives a stringent account that the bottleneck of the downstream should be worked out first, prior to the Fast Track intervention being discussed as operationally complete.

7. Recommendations

The airport management is recommended based on the evidence provided by the simulation and statistical analysis, to include the following recommendations:

Recommendation 1: Implement the Fast Track Lane Immediately

The service standard of 15 minutes is violated with the as-is system in a constant and serious manner, with the Regular Check-in queue time being more than 1,400 minutes during all 10 replications. These are minimised to about 3 minutes, a 99.8% reduction by Fast Track intervention, and the paired CI [965.40, 1063.08] indicates the value is statistically significant at the 95% level. This is a cost-effective solution because the effort to deploy the resources is conditional (activated when the wait in the queue is more than 15 minutes), so that this can be deployed only when it is necessary to operate. The implementation should be considered urgent.

Recommendation 2: Expand Security Capacity Concurrently

In Fast Track intervention, the pressure of the queue is taken to the security side, whose waiting time increases to about 99 minutes. This deterioration is statistically significant, as confirmed by the paired CI of Security Queue time. The implementation of the Fast Track should therefore be accompanied by airport management to work on security capacity. It is suggested that a follow-up simulation study should be conducted to identify how many more X-ray lanes are necessary to have a security queue time of less than 15 minutes according to Scenario 2. This is in line with the Theory of Constraints principle: optimising the capacity of bottlenecks without considering the capacity of the downstream does not increase the overall performance of the system (Goldratt, 1990).

Recommendation 3: Adopt Dynamic, Demand-Responsive Resource Allocation

The Scenario 2 example to use the value of the threshold-based deployment of resources is its conditional routing logic. Instead of a permanent Fast Track lane that is open at any given time and regardless of the demand, an adaptable staffing model with real-time queue monitoring equipment would balance the labour expenses and the standard of services provided (de Neufville and Odoni, 2013). The timing of the shifts must be scheduled with the busiest arrival times to ensure that the Fast Track agents are available at the time most are required.

Recommendation 4: Conduct Further Simulation Studies

The paper has focused on one intervention so as to be clear. Simultaneous optimisation of check-in and security operations should be considered as the future research direction, and can be studied with a factorial experiment in Simul8. These results would be further supported by sensitivity analysis under peak-hour and seasonal demand conditions that would assist the airport to prepare to experience a demand surge.

8. Reflection

The end-to-end experience of the DES project lifecycle, including conceptual modelling down to statistically supported recommendations, was offered by the simulation project as a valuable experience. A number of implications on methodology and analysis came about.

Among the aspects that were more technically challenging was the determination of the warm-up period with the help of Welch. It involved the generation of numerous replicates, time-series difference statistics calculation, a moving average, and interpretation of the stability point, which proved to be a demanding and delicate task that revealed the sensitivity of steady-state inferences to initialisation bias. As Robinson (2014) warns, the necessary time of warm-up can be overestimated by the practitioner; that is what this particular project helped to reaffirm the importance of evidence-based determination instead of clinging to the predetermined rules or intuition.

The most educative was the development of the security bottleneck in Scenario 2. This demonstrates the inimitable ability of simulation to expose non-obvious system behaviours not seen when queuing each queue individually or when applying intuition. Scenario 2 would be considered an unqualified victory as assessed on the check-in queue alone. The whole-system perspective revealed the performance trade-off, turning what seemed at first a simple 'add a lane' suggestion into more complicated a system-wide improvement programme. It coincides with the fact that the simulation is most useful related to the situations in which the interactions between the systems are most non-linear and counterintuitive (Shannon, 1998).

The presentation of two distinct paired CIs (one of the primary KPI (Regular Check-in queue)) and the other of the downstream security effect was a methodological decision. This gives a more given and sincere statistical image than having one CI and shows that the intervention, even being effective upon its claimed goal, leaves a new issue to be dealt with by management. Matched replications used across all of them minimise between-replication variance and increase the accuracy of estimation over independent CI comparison (Law, 2015).

There are some limitations that should be admitted. Distribution of service time is considered constant over the period of simulation; in reality, this is a variation with time of day, staff fatigue, and seasonal demand. The Fast Track activation logic is a simplified model of decision-making in the real world, which is probably gradual and has a number of operational factors. This set of limitations indicates that the simulation would otherwise contain good directional evidence, but piloting in reality and additional refinement would have to be done before it can be implemented on a full scale.

Comprehensively, the practical value of DES as a management decision-support tool was evidenced in this project, which is the ability to conduct evidence-based exploration of operational alternatives under a risk-free setting, for which no live experimentation would have been feasible.

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